Weekly Progress Report

Name: Devika Polavarapu

Domain: Data Science and Machine Learning

Date of submission: April 27, 2026

Week Ending: 02

I. Overview:

This week, the primary focus was on completing machine learning projects using real datasets and improving understanding of model evaluation and performance comparison. Additionally, efforts were made to strengthen implementation skills and project execution.

II. Achievements:

1. USC_TIA Familiarization:
   - Improved understanding of USC_TIA workflow and submission requirements.
   - Successfully managed project execution and report preparation

2. Python Project Contributions:

Name of the project:- Traffic Forecasting, Crop Production Prediction

- Completed traffic forecasting project using real dataset and implemented time-based feature engineering and model training.
- Completed crop production prediction project by performing data preprocessing, encoding, and applying machine learning models.
- Evaluated model performance using RMSE and MAE and compared results between models.

3.Learning Python:

- Strengthened practical knowledge of pandas, numpy, scikit-learn, matplotlib, and seaborn.

- Applied machine learning concepts to build complete end-to-end pipelines.

III. Challenges:

1. USC_TIA Integration:

- Faced challenges in aligning submission strictly with required format.
- Resolved by carefully reviewing instructions and correcting formatting issues.

2. Python Project Complexity:

- Encountered difficulty in understanding model evaluation metrics and feature impact.
- Addressed by analyzing outputs and comparing different model performances.

IV. Learning Resources:

1. USC_TIA Documentation:

- Utilized USC_TIA documentation to understand submission guidelines and workflow.
- Referred to online tutorials for resolving issues.

2. Python Learning Resources:

- Explored scikit-learn and pandas documentation for better implementation.
- Learned through hands-on practice with real datasets.

V. Next Week's Goals:

1. USC_TIA Enhancement:

- Improve efficiency in handling project submissions and workflow.

2.Python Project Development:

- Explore advanced models and improve prediction accuracy.
- Work on better visualization and analysis of results.

VI. Additional Comments:

“Gained confidence in implementing complete machine learning projects and understanding real-world data processing.”